

Exploratory Data Analysis: Subiculum Literature Dataset

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1 Summary

This EDA examines 59 years of subiculum research literature extracted from PubMed. The dataset contains 3,576 papers spanning from 1966 to 2025, with 15,603 unique authors publishing across 593 journals. The analysis reveals significant growth in subiculum research, particularly after 2010, with strong representation of neuroimaging, Alzheimer’s disease, and neuroanatomy studies.

Key Findings:

- Research has accelerated dramatically: 39% of all papers published in the last decade
- The Journal of Neuroscience and Brain Research dominate the publication landscape
- 91% of papers include MeSH term annotations enabling semantic analysis
- Citation enrichment achieved 80% coverage through CrossRef/Semantic Scholar integration
- Neuroanatomy and Alzheimer’s disease represent the largest research themes

2 Dataset Overview

2.1 Coverage Statistics

The dataset shows excellent metadata completeness, with nearly all papers including titles (100%), abstracts (98.5%), and DOIs (95.7%). This high quality enables future text mining and semantic analysis.

Table 1: Dataset coverage statistics

Metric	Value	Percentage
Total Papers	3,576	100%
Year Range	1966 – 2025	59 years
Papers with Abstracts	3,521	98.5%
Papers with DOI	3,423	95.7%
Papers with PMC ID	1,419	39.7%
Average Text Length	1,621 chars	—

3 Temporal Trends

3.1 Publication Growth

Subiculum research has grown substantially over six decades. The 2010s marked a peak in publication volume with 904 papers (25% of the entire dataset). The slight decline in the 2020s reflects incomplete data for 2025-2029 and COVID-19 impacts.

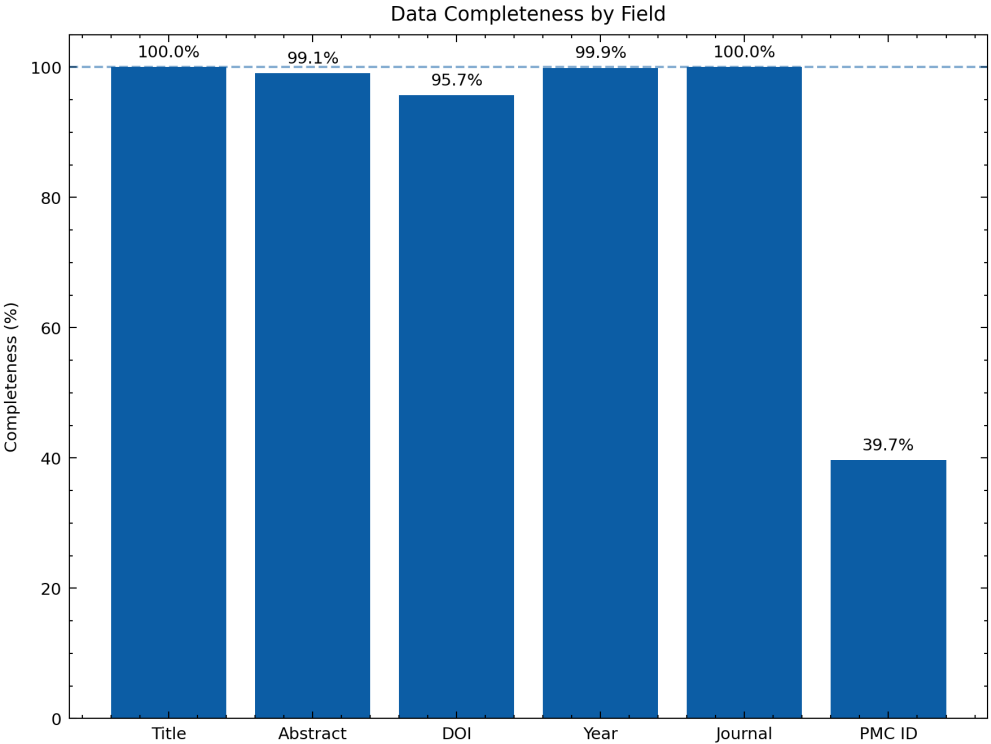


Figure 1: Data completeness by field showing high quality metadata across core bibliographic fields.

Table 2: Publication volume by decade

Decade	Papers	Growth from Previous
1960s	1	—
1970s	55	+5400%
1980s	358	+551%
1990s	717	+100%
2000s	667	-7%
2010s	904	+36%
2020s	870	-4%

3.2 Recent Acceleration

The last 10 years (2015-2025) account for 1,407 papers (39% of total), with an average of 128 papers per year. The 5-year moving average shows sustained growth, peaking at approximately 150 papers per year in 2020-2024.

Key Observation: The 2023-2025 period shows the highest sustained publication rate in subiculum research history (173-174 papers per year).

4 Journal Distribution

4.1 Top Publishing Venues

Subiculum research is published across 593 unique journals, with the top 10 accounting for 31.9% of all papers.

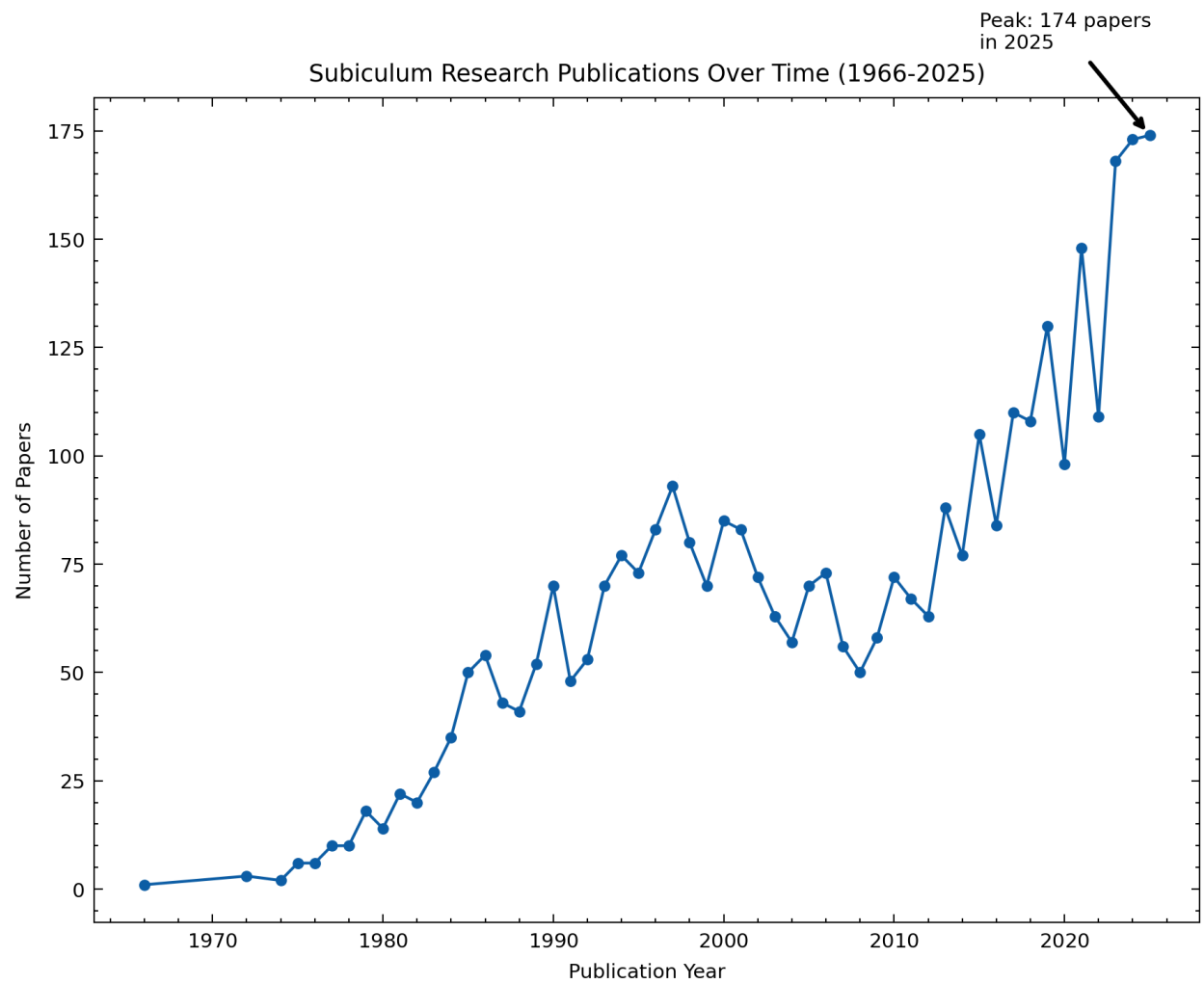


Figure 2: Annual publication counts showing sustained growth from 1966 to 2025, with peak years in 2020-2024.

Table 3: Top 10 journals by paper count with PMC access rates

Rank	Journal	Papers	PMC Access
1	Brain Research	218	4.1%
2	Journal of Comparative Neurology	212	10.4%
3	Neuroscience	166	7.2%
4	Hippocampus	128	28.9%
5	Journal of Neuroscience	122	100%
6	European Journal of Neuroscience	65	16.9%
7	Neuroscience Letters	62	1.6%
8	Acta Neuropathologica	57	15.8%
9	Behavioural Brain Research	57	12.3%
10	NeuroImage	54	44.4%

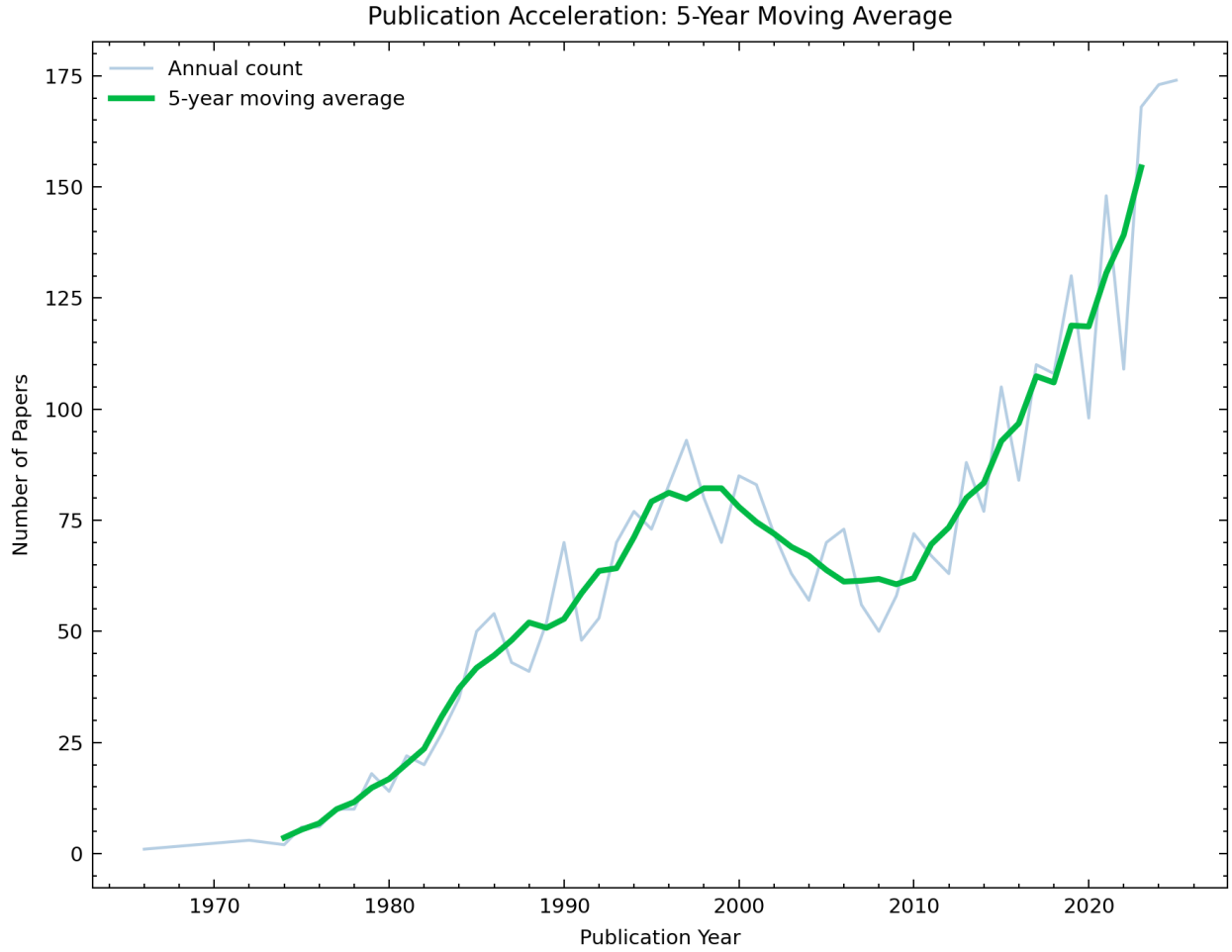


Figure 3: Publication acceleration shown through 5-year moving average, revealing steady growth with recent peak.

5 Author Contributions

5.1 Most Prolific Authors

The analysis identified 15,603 unique authors contributing to subiculum research.

Table 4: Top 5 most prolific authors in subiculum research

Rank	Author	Papers	Career Span	Years Active
1	Aggleton, John P	27	2002–2024	22 years
2	Witter, M P	25	1982–2020	38 years
3	Heinemann, U	24	1986–2015	29 years
4	Thompson, Paul M	22	2003–2023	20 years
5	Grace, Anthony A	20	2005–2022	17 years

Note: 83.3% (12,998 out of 15,603 authors) appear on only one paper in the database.

Key insights:

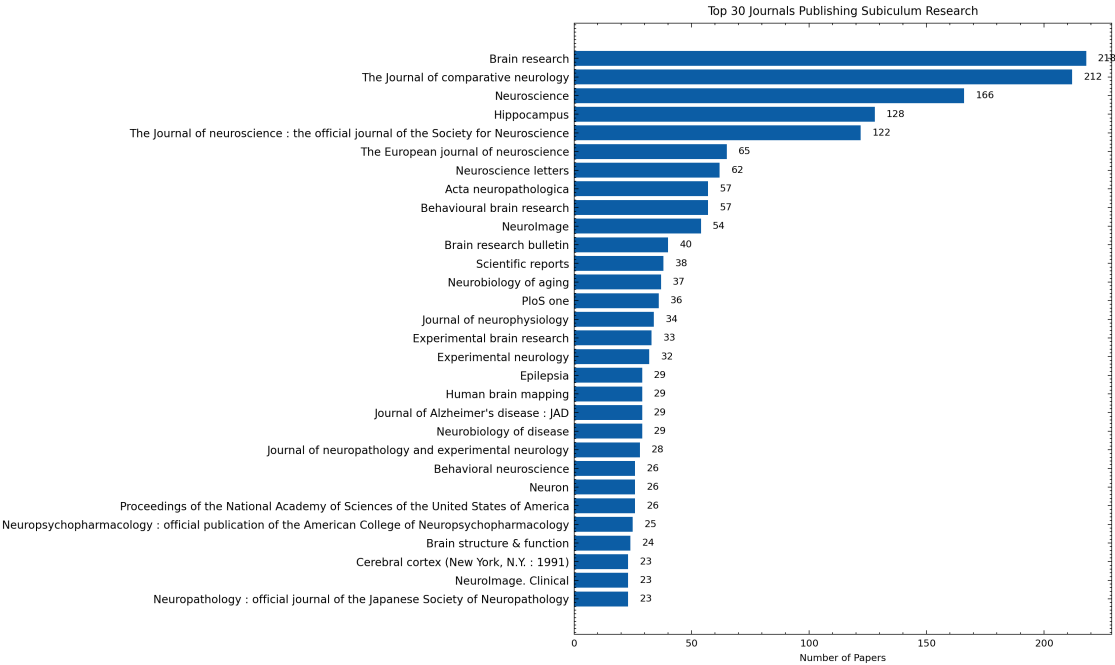


Figure 4: Top 30 journals publishing subiculum research, showing strong concentration in core neuroscience venues.

Table 5: Author contribution tiers

Author Tier	Count	% of Total
1 paper only	12,998	83.3%
2–5 papers	2,439	15.6%
6–10 papers	144	0.9%
11+ papers (core)	22	0.1%

- Core contributors: Only 22 authors (0.1%) have published 11+ papers on subiculum
- Regular contributors: 144 authors (0.9%) have 6–10 papers (see Appendix A)
- Long tail: A vast majority (83%) are “one-time” contributors

5.2 Collaboration Patterns

- Single-author papers: 171 (4.8%)
- Multi-author papers: 3,404 (95.2%)
- Average authors per paper: 4.6 (increasing over time)
- Largest collaboration: 332 authors on a single paper

Collaboration has increased steadily, with the average number of authors per paper rising from approximately 2 in the 1970s to approximately 6 in the 2020s.

6 Citation Network Analysis

6.1 Citation Coverage

Citation enrichment via CrossRef and Semantic Scholar achieved:

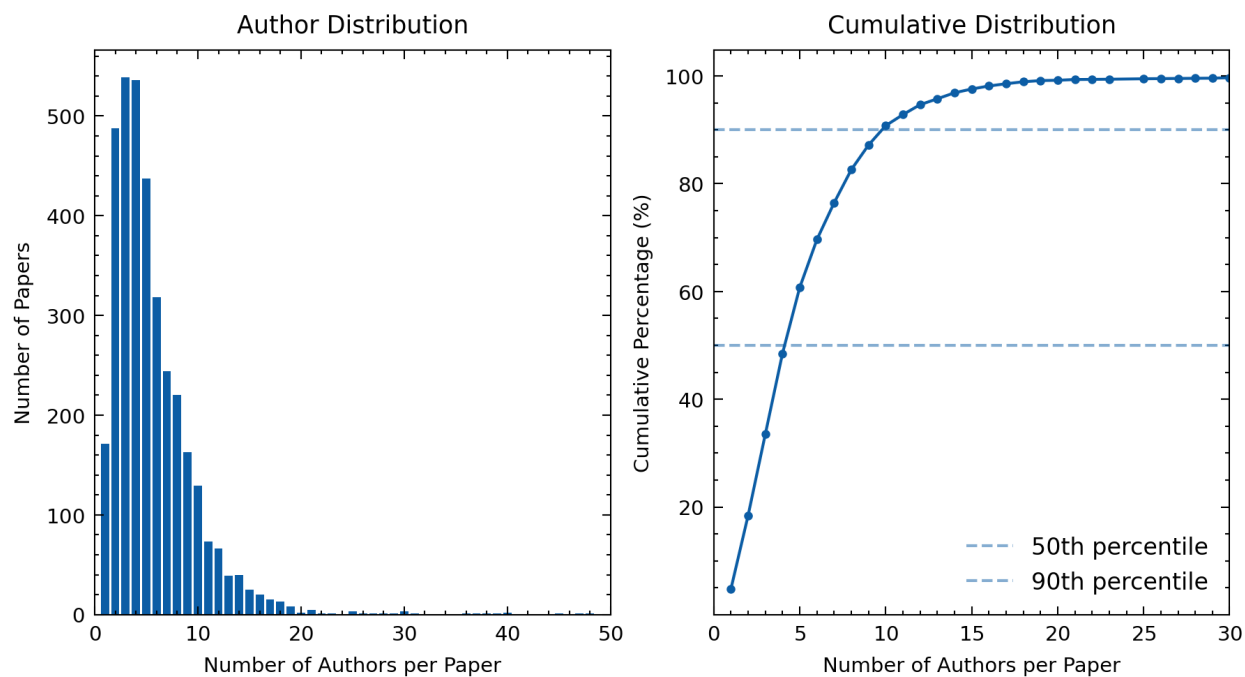


Figure 5: Distribution of author counts per paper showing predominance of small collaborative teams (left) and cumulative distribution (right).

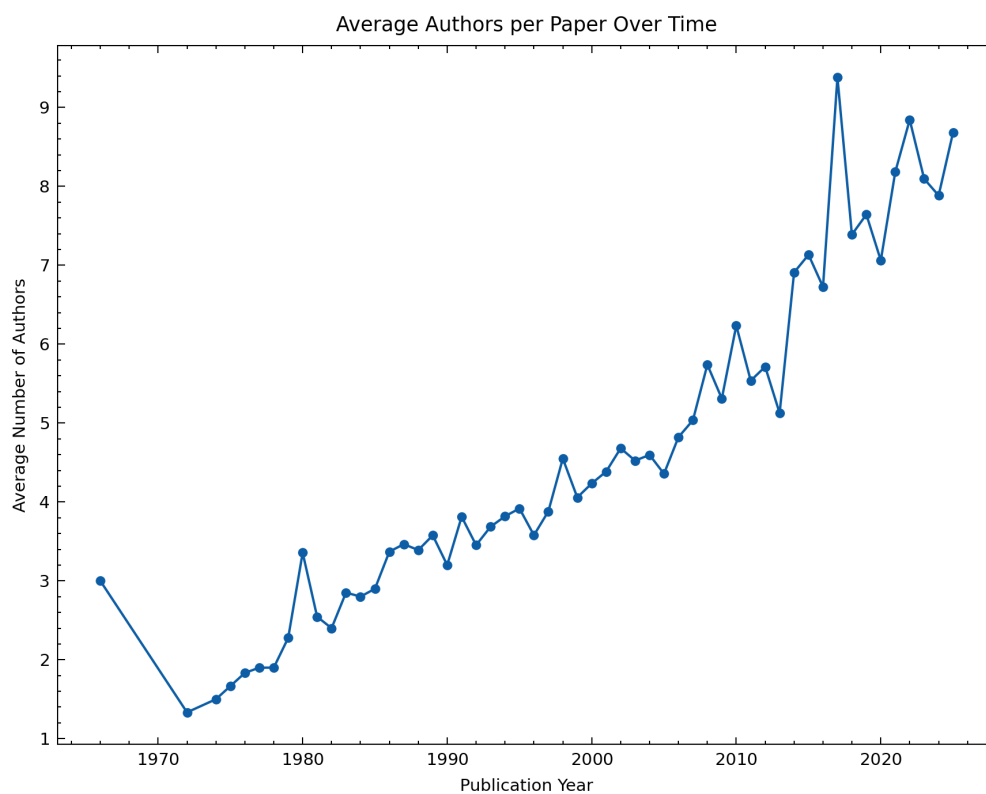


Figure 6: Average number of authors per paper over time, showing steady increase in collaboration.

- 2,874 papers with reference lists (80.4%)
- 98,445 total citation links
- Average 34.3 references per paper (for papers with citations)

6.2 Most Influential Papers

Table 6: Top 5 most cited papers in the dataset

PMID	Title (Year)	Cited	Journal
65364	An autoradiographic study... (1977)	220	J. Comp. Neurol.
11240210	The subiculum: a review... (2001)	160	Prog. Neurobiol.
16185252	The subiculum: what it does... (2005)	114	J. Anatomy
1727001	Organization of CA1 projections... (1991)	110	Hippocampus
3683859	Organization of projections... (1987)	109	Neuroscience

Classic paper: The most cited work (PMID 65364, 1977) remains foundational after 48 years, demonstrating the enduring importance of early neuroanatomical studies.

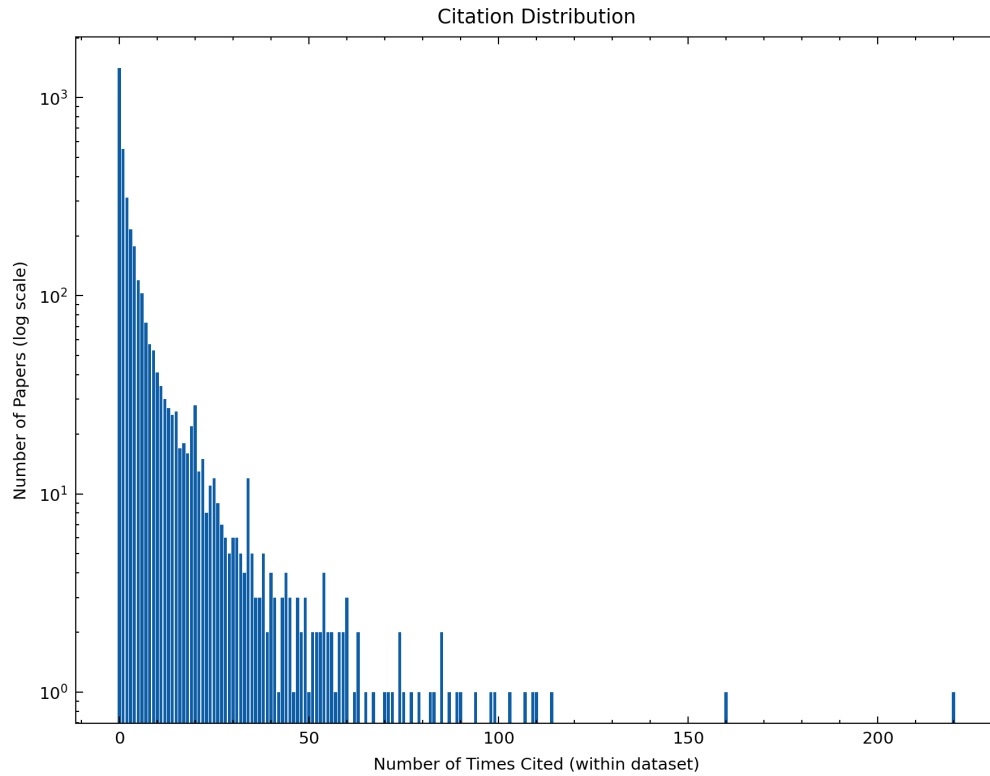


Figure 7: Citation distribution showing typical long-tail pattern with few highly-cited papers and many with few citations.

6.3 Citation Coverage by Era

Recent papers have excellent citation coverage, while pre-2000 papers show gaps that may require additional enrichment from historical databases.

Table 7: Citation coverage by decade

Decade	Citation Coverage
1960s–1970s	45.5%
1980s–1990s	63.5%
2000s	72.7%
2010s	92.5%
2020s	97.4%

7 Research Themes

7.1 MeSH Term Analysis

Coverage: 91.2% of papers (3,262/3,576) have MeSH annotations.

Top 10 MeSH terms reveal core research domains, with hippocampus and animal models dominating, alongside significant human and clinical research focus on aging and Alzheimer’s disease.

Table 8: Top 10 MeSH terms in the dataset

MeSH Term	Papers	% of Dataset
Hippocampus	2,330	65.2%
Animals	2,118	59.2%
Male	2,059	57.6%
Female	1,243	34.8%
Humans	1,379	38.6%
Rats	1,287	36.0%
Neurons	764	21.4%
Aged	594	16.6%
Brain	593	16.6%
MRI	560	15.7%
Alzheimer Disease	460	12.9%

Key Insight: The predominance of animal models (59%) alongside human studies (39%) reflects the field’s dual focus on basic neuroscience and clinical translation. Alzheimer’s disease represents the largest disease-specific research area.

7.2 Author Keywords

Coverage: 35% of papers (1,250/3,576) include author-assigned keywords.

Top author-provided keywords show research focus areas: hippocampus (305 papers), subiculum (146 papers), Alzheimer’s disease (105 papers), MRI (67 papers), and hippocampal subfields (57 papers).

Note: MeSH terms provide more comprehensive semantic coverage (91%) than author keywords (35%), making them preferable for topic modeling.

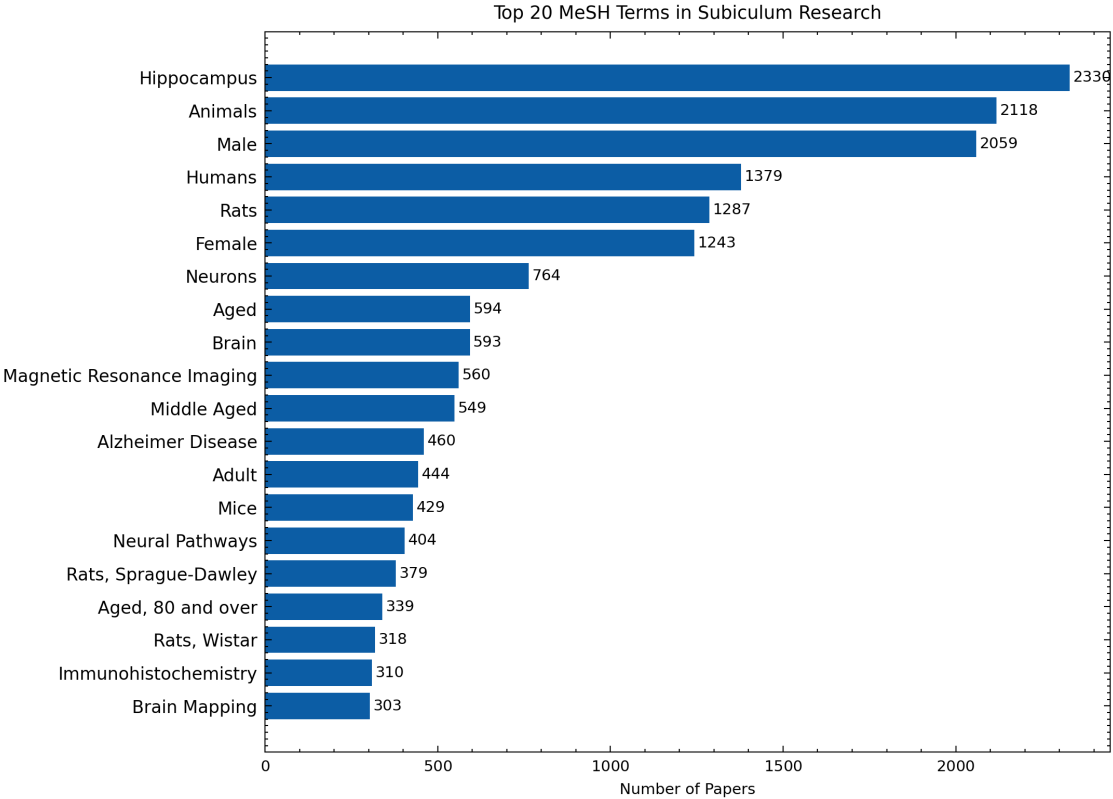


Figure 8: Top 20 MeSH terms showing dominance of neuroanatomical and disease-focused research.

8 Publication Types

The dataset is dominated by primary research articles, with over 50% of papers acknowledging external research support and NIH funding appearing in approximately 28% of papers.

Table 9: Distribution of publication types

Publication Type	Count	% of Total
Journal Article	3,569	99.8%
Research Support, Non-U.S. Gov’t	1,853	51.8%
Research Support, U.S. Gov’t, P.H.S.	510	14.3%
Research Support, N.I.H., Extramural	485	13.6%
Comparative Study	293	8.2%
Review	136	3.8%
Case Reports	51	1.4%
Preprint	25	0.7%

9 Data Quality Assessment

The dataset exhibits excellent metadata quality, with greater than 95% completeness for core bibliographic fields.

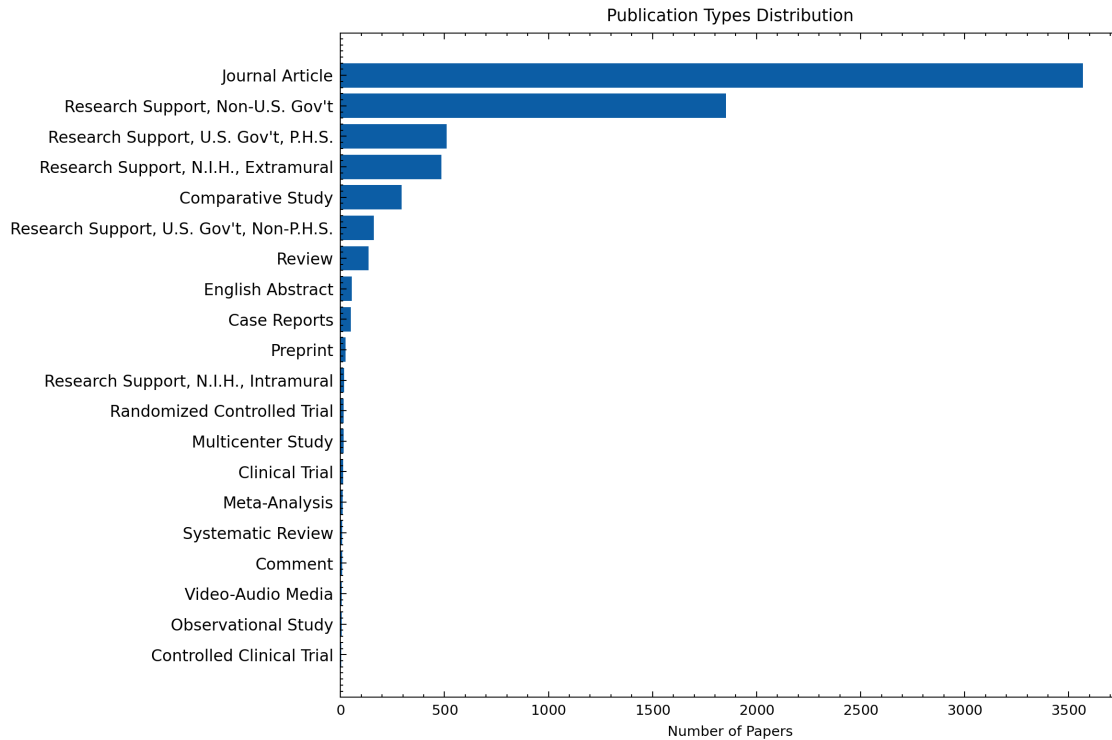


Figure 9: Publication type distribution showing dominance of research articles with significant funding support.

Table 10: Metadata completeness summary

Field	Completeness
Title	100%
Journal	100%
Year	99.9%
Abstract	99.1%
DOI	95.7%
PMC ID	39.7%

10 Key Takeaways

10.1 Research Landscape

1. **Accelerating field:** 39% of all subiculum research published in last 10 years
2. **Interdisciplinary:** Spans neuroscience, neuroimaging, psychiatry, and aging research
3. **Well-documented:** 91% MeSH coverage enables semantic analysis
4. **Collaborative:** Average 4.6 authors per paper, increasing over time

10.2 Dominant Themes

1. **Neuroanatomy & Connectivity:** Foundation of the field (1970s–1990s classics)
2. **Alzheimer's Disease:** Largest clinical application (460 papers, 12.9%)
3. **Neuroimaging:** Methodological evolution (MRI: 560 papers, 15.7%)
4. **Animal Models:** Predominantly rodent studies (59% of papers)

10.3 Data Strengths

- Comprehensive temporal coverage (59 years)
- High metadata quality (>95% completeness)
- Rich semantic annotations (91% MeSH terms)
- Strong citation network (80% coverage)

10.4 Data Limitations

- Pre-2000 citation data incomplete (45–65% coverage)
- Author keywords sparse (35% of papers)
- Limited to PubMed-indexed literature (excludes books, dissertations)

11 Future Directions

11.1 For Literature Review

- Focus on 2010+ papers for best citation coverage
- Prioritize MeSH terms over author keywords for topic modeling
- Consider top 10 journals for comprehensive coverage (32% of literature)

11.2 For Citation Analysis

- Supplement pre-2000 papers with Web of Science or Scopus
- Strong foundation in comparative neuroanatomy

11.3 For Bibliometric Analysis

- Collaboration networks show increasing complexity over time
- NIH-funded research represents core of U.S. contributions
- European Journal of Neuroscience represents major international venue

A Regular Contributors (6–10 Papers)

This appendix provides a complete list of all 144 authors who have published 6–10 papers on subiculum research. These regular contributors represent 0.9% of all authors but have sustained engagement with the field over multiple years.

Table 11: Complete list of regular contributors (6–10 papers)

Author	Papers	First Pub	Latest Pub	Years Active
Wozny, Christian	10	2005	2025	20
Raz, Naftali	10	2011	2025	14
Wang, Yi	10	2012	2025	13
Vann, Seralynne D	10	2005	2022	17
Spruston, Nelson	10	2003	2019	16
Lévesque, Maxime	10	2011	2018	7
Miles, Richard	10	2002	2016	14
Rawlins, J N	10	1986	2001	15
Hyman, B T	10	1984	1998	14
Jarrard, L E	10	1978	1997	19
Tohyama, M	10	1980	1993	13
de Flores, Robin	9	2015	2025	10
Ding, Song-Lin	9	2007	2025	18
Insausti, Ricardo	9	2010	2025	15
Fei, Fan	9	2021	2025	4
Mukherjee, Jogeshwar	9	2006	2025	19
Chen, Zhong	9	2019	2025	6
Chételat, Gaël	9	2010	2025	15
Chakravarty, M Mallar	9	2013	2025	12
Jack, Clifford R	9	2006	2024	18
Whitwell, Jennifer L	9	2006	2024	18
Amin, Eman	9	2003	2024	21
Wisse, Laura E M	9	2014	2023	9
Sperk, Günther	9	2003	2022	19
Kutty, Bindu M	9	2003	2022	19
Toga, Arthur W	9	2006	2017	11
O'Mara, S M	9	1998	2017	19
Lopes da Silva, F H	9	1982	2009	27
Wouterlood, F G	9	1990	2008	18
Totterdell, S	9	1986	2005	19
Hof, P R	9	1987	2002	15
West, M J	9	1984	2001	17
Gloveli, T	9	1994	2001	7
Wyss, J M	9	1983	1999	16
Yanagihara, T	9	1985	1995	10
Chan-Palay, V	9	1981	1988	7
Uemura, E	9	1979	1985	6

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Author	Papers	First Pub	Latest Pub	Years Active
Moon, Minho	8	2012	2025	13
Huang, Yushan	8	2015	2025	10
Petersen, Ronald C	8	2006	2024	18
Josephs, Keith A	8	2006	2024	18
Drexel, Meinrad	8	2012	2023	11
Schmitz, Dietmar	8	2008	2021	13
Miller, Michael I	8	2003	2017	14
Csernansky, John G	8	2003	2016	13
Stewart, M	8	1997	2008	11
Kawasaki, H	8	1996	2007	11
de Leon, M J	8	1990	2002	12
Harrison, P J	8	1990	2001	11
Feldon, J	8	1986	2000	14
Price, J L	8	1974	1999	25
McEwen, B S	8	1982	1998	16
Cotman, C W	8	1986	1997	11
Bossert, Jennifer M	7	2013	2025	12
Landeau, Brigitte	7	2010	2025	15
Saunders, Richard C	7	2002	2024	22
Joksimovic, Srdjan M	7	2017	2023	6
Zeineh, Michael M	7	2003	2021	18
Anderson, M	7	1998	2020	22
Herman, James P	7	2002	2018	16
Apostolova, Liana G	7	2006	2017	11
Cooper, Donald C	7	2002	2013	11
Raju, T R	7	1997	2011	14
Gonzalez-Lima, F	7	1994	2011	17
Arnold, S E	7	1991	2007	16
Pitkänen, A	7	1992	2004	12
Armstrong, D M	7	1988	2003	15
Yee, B K	7	1995	2002	7
Bouras, C	7	1994	2002	8
Morrison, J H	7	1988	2002	14
Commins, S	7	1998	2001	3
Gray, J A	7	1985	2001	16
Joyce, J N	7	1989	2000	11
Geneser, F A	7	1987	1997	10
Zimmer, J	7	1978	1996	18
Gessi, T	7	1979	1995	16
Siesjö, B K	7	1984	1995	11
Bennett, David A	6	2003	2025	22
Riggins, Tracy	6	2018	2025	7
Ofen, Noa	6	2016	2025	9
Liang, Christopher	6	2023	2025	2

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Author	Papers	First Pub	Latest Pub	Years Active
Luders, Eileen	6	2013	2025	12
Kurth, Florian	6	2013	2025	12
Xu, Cenglin	6	2017	2025	8
Chen, Zhong	6	2012	2025	13
Wang, Yi	6	2022	2025	3
Mézenge, Florence	6	2010	2025	15
Waldvogel, Henry J	6	2017	2025	8
Lever, Colin	6	2006	2025	19
Gong, Qiyong	6	2018	2025	7
Wolk, David A	6	2018	2025	7
Xu, Xiangmin	6	2014	2024	10
La Joie, Renaud	6	2010	2024	14
Petrucelli, Leonard	6	2014	2023	9
Holmes, Todd C	6	2016	2023	7
Tesic, Vesna	6	2019	2023	4
Todorovic, Slobodan M	6	2011	2023	12
Veltman, Dick J	6	2015	2023	8
Honda, Yoshiko	6	2004	2022	18
Bender, Andrew R	6	2013	2022	9
Yoshida, Mari	6	2009	2022	13
Das, Sandhitsu R	6	2010	2021	11
Trojanowski, J Q	6	1991	2021	30
Tang, Xiaoying	6	2015	2020	5
Kinnavane, Lisa	6	2016	2020	4
Stark, Craig E L	6	2008	2020	12
Dillingham, Christopher M	6	2015	2019	4
Frisoni, Giovanni B	6	2012	2019	7
Somogyi, Peter	6	2002	2018	16
Argiolas, Antonio	6	2009	2017	8
Melis, Maria Rosaria	6	2009	2017	8
Gigg, John	6	2006	2016	10
Wright, Nicholas F	6	2010	2016	6
Ekstrom, Arne D	6	2010	2015	5
Bookheimer, Susan Y	6	2003	2015	12
Baulac, Michel	6	2002	2015	13
Akiyama, Haruhiko	6	2002	2014	12
Thom, M	6	1997	2009	12
Zilles, K	6	1985	2005	20
Eichenbaum, H	6	1983	2005	22
Bowery, N G	6	1986	2003	17
Ikonomic, M D	6	1995	2003	8
van Groen, T	6	1986	2003	17
Wegiel, J	6	1993	2002	9
Vinogradova, O S	6	1976	2001	25

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Table 11 – *Continued from previous page*

Author	Papers	First Pub	Latest Pub	Years Active
Yang, C R	6	1984	2001	17
Naber, P A	6	1998	2001	3
Ferrer, I	6	1993	2001	8
Greene, J R	6	1996	2001	5
Winblad, B	6	1997	2000	3
Bobinski, M	6	1993	2000	7
Wisniewski, H M	6	1993	2000	7
Berger, B	6	1985	1999	14
Moore, R Y	6	1977	1997	20
Slomianka, L	6	1987	1997	10
Hoyer, D	6	1994	1996	2
Gasbarri, A	6	1989	1996	7
Pacitti, C	6	1989	1996	7
Braak, H	6	1972	1996	24
Buzsáki, G	6	1986	1994	8
Vanderhaeghen, J J	6	1989	1993	4
Loy, R	6	1979	1988	9
Rainbow, T C	6	1982	1985	3
Siegel, A	6	1977	1981	4

Notable contributors:

- Trojanowski, J Q has the longest career span: 30 years (1991–2021)
- Vinogradova, O S started earliest (1976) with 25 years of contributions
- Braak, H published from 1972–1996 (24 years) – important early Alzheimer’s researcher
- Lopes da Silva, F H maintained 27 years of contributions (1982–2009)

B Data Files Reference

All analysis outputs are saved as CSV files in the `outputs/` directory with timestamp 2025-12-06. The following files are referenced throughout this report:

B.1 Core Dataset Files

- `dataset_overview.csv` – Summary statistics of the complete dataset
- `data_completeness.csv` – Field-level completeness metrics

B.2 Temporal Analysis

- `papers_by_year.csv` – Annual publication counts (1966–2025)
- `papers_by_decade.csv` – Decadal aggregation of publications

B.3 Publication Venues

- `top_journals.csv` – Journal rankings by publication count
- `publication_types.csv` – Distribution of article types (research, review, etc.)

B.4 Author Contributions

- `top_100_authors.csv` – Most prolific authors in subiculum research

B.5 Citation Metrics

- `most_cited_papers.csv` – Highly cited papers in the dataset
- `citation_distribution.csv` – Distribution of citation counts across papers

B.6 Research Themes

- `top_mesh_terms.csv` – Most frequent MeSH terms
- `top_keywords.csv` – Most frequent author-assigned keywords

C San Diego Institutional Contributions

San Diego area institutions represent less than 1% of the dataset (33/3,576 papers, 0.9%).

Table 12: San Diego area institutional contributions

Institution	Papers	Authors
UCSD	26	30+
Salk Institute	7	8
Neurosciences Institute	2	5

UCSD Top Contributors: Douglas Nitz (Cognitive Science) is the most prolific local contributor with 5 papers (2017-2025), focusing on spatial cognition and hippocampal subfield function. Recent work includes Alzheimer’s disease research (Psychiatry), spatial coding (Cognitive Science), and developmental neuroscience (Neurosciences Department).

Salk Institute: Historical contributions in 1980s-1990s, with recent 2014-2015 systems neurobiology work. Notable: David Amaral (now at UC Davis) published from Salk in 1991; Edward Callaway is a prominent systems neuroscientist.

C.1 Complete List of San Diego Papers

Table 13: All papers with San Diego institutional affiliations (n=40)

PMID	Title	Year	Authors
39511453	Projection-specific circuits of retrosplenial cortex with differential contributions to spatial cognition.	2025	Lin X, et al.
40247152	Sex- and region-dependent neuroinflammation in Alzheimer’s disease.	2025	Acosta-Martínez M, et al.
40442306	Individual differences in medial temporal lobe functional network architecture predict the capacity for sleep-related consolidation of emotional memories.	2025	Chappel-Farley M, et al.
41139612	Memory Performance and Hippocampal Volumes in Healthy Weight Children at High- and Low-Familial Risk for Obesity.	2025	Dratva M, et al.
38448584	Subicular neurons encode concave and convex geometries.	2024	Sun Y, Nitz D, Xu X
38600385	Cell-type-resolved mosaicism reveals clonal dynamics of the human forebrain.	2024	Chung C, Yang X, Hevner R

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PMID	Title	Year	Authors
37035460	Differential involvement of hippocampal subfields in the relationship between Alzheimer's pathology and memory interference.	2023	Adams J, et al.
36121513	Molecular diversity and phenotypic pleiotropy of ancient genomic regulatory loci derived from human endogenous retrovirus.	2022	Glinsky G
33232797	Hippocampal subfield abnormalities and memory functioning in children with fetal alcohol Spectrum disorders.	2021	Roediger D, et al.
31548723	CA1-projecting subiculum neurons facilitate object-place learning.	2019	Sun Y, Jin S, Lin X
29387780	Opposing and Complementary Topographic Connectivity Gradients in Hippocampal CA1 Inputs.	2018	Sun Y, Nitz D, Holmes T
27991899	Subiculum neurons map the current axis of travel.	2017	Olson J, et al.
28098162	Novel genetic loci associated with hippocampal volume.	2017	Hibar D, et al.
25127742	In vivo hippocampal subfield volumes in schizophrenia and bipolar disorder.	2015	Haukvik U, et al.
26010947	Evidence for limbic paraventricular hypothalamic inhibitory network in HPA axis adaptations to stress.	2015	Radley J, Sawchenko P
26401700	APOE Affects Volume and Shape of Amygdala and Hippocampus in MCI and Alzheimer's Disease.	2015	Holland D, Dale A, et al.
24656815	Cell-type-specific circuit connectivity of hippocampal CA1 revealed through Cre-dependent rabies tracing.	2014	Choi J, Callaway E
21831652	Nicotinic 42 receptor imaging agents. Synthesis and evaluation of 18F-nifzetidine.	2011	Pichika R, et al.
19854244	Pathways from ventral hippocampus and caudal amygdala to forebrain sensorimotor gating regions.	2010	Miller E, et al.
17101872	The anatomy of amnesia: neurohistological analysis of three new cases.	2006	Gold J, et al.
12040072	Anterograde amnesia and temporally graded retrograde amnesia after hippocampus and subiculum lesions.	2002	Clark R, et al.

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PMID	Title	Year	Authors
12353223	Estrogen receptor-beta colocalizes with parvalbumin-labeled inhibitory neurons in cortex and hippocampus.	2002	Blurton-Jones M, et al.
11306005	Regulation of sensorimotor gating in rats by hippocampal NMDA: anatomical localization.	2001	Swerdlow N, et al.
10627621	Impaired recognition memory in monkeys after damage limited to the hippocampal region.	2000	Zola S, et al.
10051156	Nucleus accumbens projections involved in D1-like dopamine receptor-induced locomotor activity.	1999	Hooks M, et al.
9884323	Brainstem GABAergic projection neurons express beta 2 nicotinic acetylcholine receptors.	1999	Gallagher D, et al.
8878739	Hippocampal regulation of dopaminergic responses to stress: role of the ventral subiculum.	1996	Herman J, et al.
8878738	Blockade of hippocampal NMDA receptors prevents stress-induced changes in dopamine function.	1996	Shoemaker W, et al.
8878736	Swim stress-induced changes in extracellular GABA in the locus coeruleus of the rat.	1996	Mana M, et al.
8586476	Are subicular outputs affected in Alzheimer's disease?	1995	Van Hoesen G, et al.
7791146	Correlations between Golgi neurons, choline acetyltransferase, and NADPH-diaphorase histochemistry.	1995	Jaarsma D, et al.
8159293	Entorhinal and perirhinal projections to hippocampus: topography and laminar organization.	1993	Naber P, et al.
2027854	Corticosterone decreases tyrosine hydroxylase but not choline acetyltransferase in specific regions.	1991	Troxler T, et al.
1828213	Ultrastructure of GABA, parvalbumin, and calbindin-D28k immunoreactive neurons in monkey hippocampus.	1991	Sloviter R, et al.
1828212	Concurrent visualization of GABA and parvalbumin in hippocampal neurons.	1991	Sloviter R, et al.
1828211	Density and distribution of hippocampal neurotransmitter receptors in Alzheimer's disease.	1991	Jansen K, et al.

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Table 13 – *Continued from previous page*

PMID	Title	Year	Authors
1655889	Organization of multiple inputs from the amygdala to the nucleus accumbens.	1991	Wright C, et al.
1974755	Input from the retrosplenial cortex to the amygdaloid complex in the rat.	1990	Shibata H
2251286	Serotonin terminals synapse on septohippocampal neurons in the rat.	1990	Tork I, et al.
3146020	The medial septum sends GABA-immunoreactive axons to dentate gyrus and stratum oriens.	1988	Freund T, et al.

D Top 25 Most Cited Papers

The following table lists the 25 most highly cited papers in the subiculum literature dataset, based on citation data from CrossRef and Semantic Scholar APIs. These papers represent foundational contributions to subiculum research in neuroanatomy, electrophysiology, and clinical neuroscience.

Table 14: Top 25 most cited papers in the dataset

PMID	Title	Year	Authors	Cites
65364	An autoradiographic study of the organization of the efferent connections of the hippocampal formation in the rat.	1977	Swanson L, Cowan W	219
11240210	The subiculum: a review of form, physiology and function.	2001	O'Mara S, et al.	159
16185252	The subiculum: what it does, what it might do, and what neuroanatomy has yet to tell us.	2005	O'Mara S	114
1727001	Organization of CA1 projections to the subiculum: a PHA-L analysis in the rat.	1991	Amaral D, Dolorfo C, Alvarez-Royo P	110
3683859	Organization of the projections from the subiculum to the ventral striatum in the rat.	1987	Witter M, et al.	109
410102	Hippocampal efferents reach widespread areas of cerebral cortex and amygdala in the rhesus monkey.	1977	Rosene D, Van Hoesen G	107
12434059	On the origin of interictal activity in human temporal lobe epilepsy in vitro.	2002	Miles R, et al.	100
1270625	Topographic organization of the projections from the entorhinal area to the hippocampal formation.	1976	Steward O	99
6474172	Alzheimer's disease: cell-specific pathology isolates the hippocampal formation.	1984	Hyman B, et al.	98
3902916	Intrinsic projections of the retrohippocampal region in the rat brain. I. The subicular complex.	1985	Köhler C	94
8158272	Spatial correlates of firing patterns of single cells in the subiculum of the freely moving rat.	1994	Sharp P, Green C	89
11067982	Resting and active properties of pyramidal neurons in subiculum and CA1 of rat hippocampus.	2000	Spruston N, et al.	87
402984	Efferent connections of the hippocampal formation in the rat.	1977	Siegel A, Meibach R	87

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Table 14 – *Continued from previous page*

PMID	Title	Year	Authors	Cites
16713659	Measurement of hippocampal subfields and age-related changes with high resolution MRI at 4T.	2007	Weiner M, et al.	85
1430328	Projections of the ventral subiculum to the amygdala, septum, and hypothalamus.	1992	Canteras N, Swanson L	85
19393282	Roles for the subiculum in spatial information processing, memory, motivation and temporal control.	2009	O'Mara S, et al.	82
7903643	Electrophysiological properties of neurons in the rat subiculum in vitro.	1993	Taube J	82
16876886	Connections of the subiculum of the rat: topography in relation to columnar and laminar organization.	2006	Witter M	79
2392570	The subiculum: cytoarchitectonically a simple structure, but hodologically complex.	1990	Witter M, Groenewegen H	77
25596463	Quantitative comparison of 21 protocols for labeling hippocampal subfields in MRI.	2015	Yushkevich P, et al.	75
23839777	Comparative anatomy of the prosubiculum, subiculum, presubiculum, postsubiculum, and parasubiculum.	2013	Ding S	74
12106063	Ibotenate Lesions of Hippocampus and/or Subiculum: Dissociating Components of Allocentric Spatial Learning.	1990	Morris R, et al.	74
1783682	Distribution of hippocampal CA1 and subicular efferents in the prefrontal cortex of the rat.	1991	Witter M, Jay T	72
19405132	Selective effect of age, Apo e4, and Alzheimer's disease on hippocampal subfields.	2009	Mueller S, Weiner M	71
1713237	Entorhinal cortex of the monkey: V. Projections to the dentate gyrus, hippocampus, and subicular complex.	1991	Witter M, Amaral D	70